

```
In [4]: # import python Libraries

import numpy as np
import pandas as pd
import matplotlib.pyplot as plt # visualizing data
%matplotlib inline
import seaborn as sns
```

```
In [17]: # import csv file
df = pd.read_csv(r'C:\Users\Harshada shirke\Downloads\Diwali Sales Data.csv', encoding= 'unic
```

```
In [18]: df.shape
```

```
Out[18]: (11251, 15)
```

```
In [19]: df.shape
```

```
Out[19]: (11251, 15)
```

```
In [5]: df.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 11251 entries, 0 to 11250
Data columns (total 15 columns):
#   Column                Non-Null Count  Dtype  
---  -
0   User_ID                11251 non-null  int64  
1   Cust_name              11251 non-null  object  
2   Product_ID            11251 non-null  object  
3   Gender                 11251 non-null  object  
4   Age Group              11251 non-null  object  
5   Age                    11251 non-null  int64  
6   Marital_Status         11251 non-null  int64  
7   State                  11251 non-null  object  
8   Zone                   11251 non-null  object  
9   Occupation             11251 non-null  object  
10  Product_Category       11251 non-null  object  
11  Orders                 11251 non-null  int64  
12  Amount                 11239 non-null  float64 
13  Status                 0 non-null      float64 
14  unnamed1               0 non-null      float64 
dtypes: float64(3), int64(4), object(8)
memory usage: 1.3+ MB
```

```
In [20]: #drop unrelated/blank columns
df.drop(['Status', 'unnamed1'], axis=1, inplace=True)
```

```
In [21]: #check for null values
pd.isnull(df).sum()
```

Out[21]:

User_ID	0
Cust_name	0
Product_ID	0
Gender	0
Age Group	0
Age	0
Marital_Status	0
State	0
Zone	0
Occupation	0
Product_Category	0
Orders	0
Amount	12

dtype: int64

In [22]:

```
# drop null values
df.dropna(inplace=True)
```

In [23]:

```
# change data type
df['Amount'] = df['Amount'].astype('int')
```

In [24]:

```
df['Amount'].dtypes
```

Out[24]:

```
dtype('int32')
```

In [25]:

```
df.columns
```

Out[25]:

```
Index(['User_ID', 'Cust_name', 'Product_ID', 'Gender', 'Age Group', 'Age',
      'Marital_Status', 'State', 'Zone', 'Occupation', 'Product_Category',
      'Orders', 'Amount'],
      dtype='object')
```

In [28]:

```
#rename column
df.rename(columns= {'Marital_Status': 'Shaadi'})
```

Out[28]:

	User_ID	Cust_name	Product_ID	Gender	Age Group	Age	Shaadi	State	Zone	Occupation
0	1002903	Sanskriti	P00125942	F	26-35	28	0	Maharashtra	Western	Healthcare
1	1000732	Kartik	P00110942	F	26-35	35	1	Andhra Pradesh	Southern	Govt
2	1001990	Bindu	P00118542	F	26-35	35	1	Uttar Pradesh	Central	Automobile
3	1001425	Sudevi	P00237842	M	0-17	16	0	Karnataka	Southern	Construction
4	1000588	Joni	P00057942	M	26-35	28	1	Gujarat	Western	Food Processing
...	...	...	...	...	...	...	...	...	...	...
11246	1000695	Manning	P00296942	M	18-25	19	1	Maharashtra	Western	Chemical
11247	1004089	Reichenbach	P00171342	M	26-35	33	0	Haryana	Northern	Healthcare
11248	1001209	Oshin	P00201342	F	36-45	40	0	Madhya Pradesh	Central	Textile
11249	1004023	Noonan	P00059442	M	36-45	37	0	Karnataka	Southern	Agriculture
11250	1002744	Brumley	P00281742	F	18-25	19	0	Maharashtra	Western	Healthcare

11239 rows × 13 columns

In [29]:

```
# describe() method returns description of the data in the DataFrame (i.e. count, mean, std,
df.describe()
```

Out[29]:

	User_ID	Age	Marital_Status	Orders	Amount
count	1.123900e+04	11239.000000	11239.000000	11239.000000	11239.000000
mean	1.003004e+06	35.410357	0.420055	2.489634	9453.610553
std	1.716039e+03	12.753866	0.493589	1.114967	5222.355168
min	1.000001e+06	12.000000	0.000000	1.000000	188.000000
25%	1.001492e+06	27.000000	0.000000	2.000000	5443.000000
50%	1.003064e+06	33.000000	0.000000	2.000000	8109.000000
75%	1.004426e+06	43.000000	1.000000	3.000000	12675.000000
max	1.006040e+06	92.000000	1.000000	4.000000	23952.000000

In [30]:

```
# use describe() for specific columns
df[['Age', 'Orders', 'Amount']].describe()
```

Out[30]:

	Age	Orders	Amount
count	11239.000000	11239.000000	11239.000000
mean	35.410357	2.489634	9453.610553
std	12.753866	1.114967	5222.355168
min	12.000000	1.000000	188.000000
25%	27.000000	2.000000	5443.000000
50%	33.000000	2.000000	8109.000000
75%	43.000000	3.000000	12675.000000
max	92.000000	4.000000	23952.000000

Exploratory Data Analysis

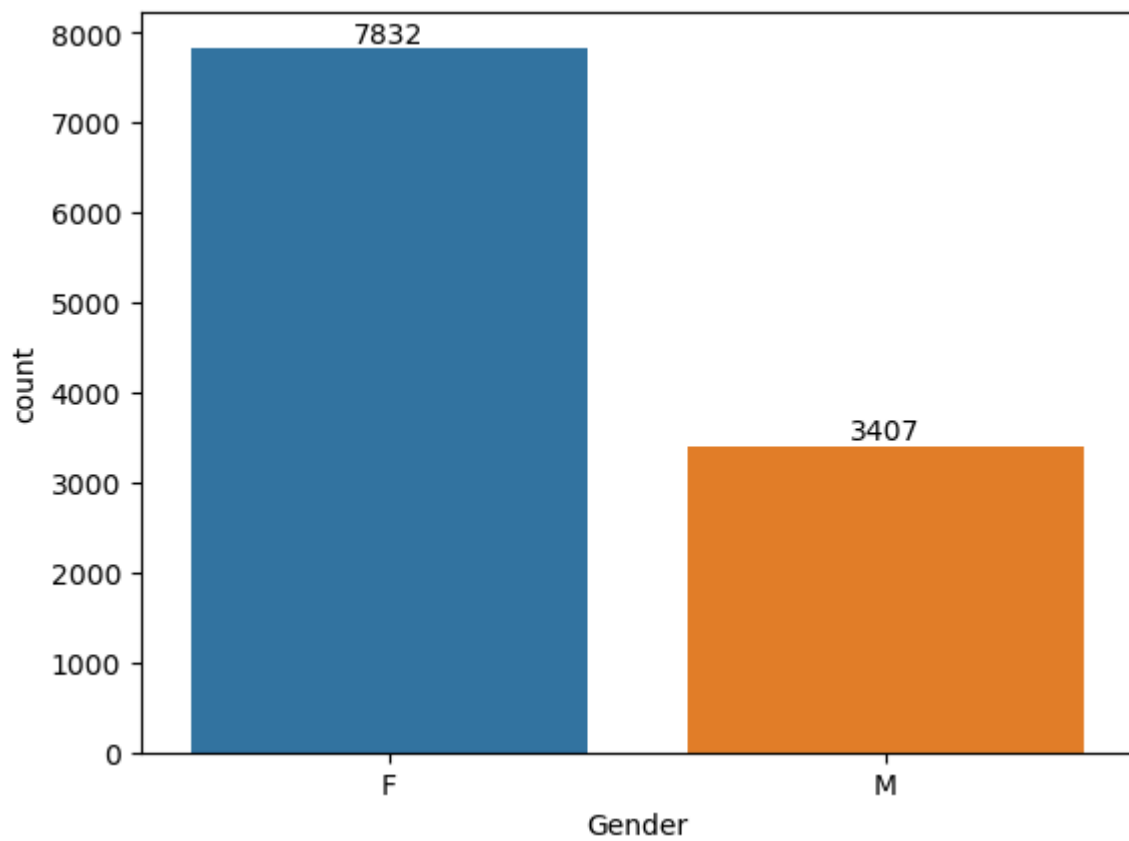
Gender

In [31]:

```
# plotting a bar chart for Gender and it's count

ax = sns.countplot(x = 'Gender', data = df)

for bars in ax.containers:
    ax.bar_label(bars)
```

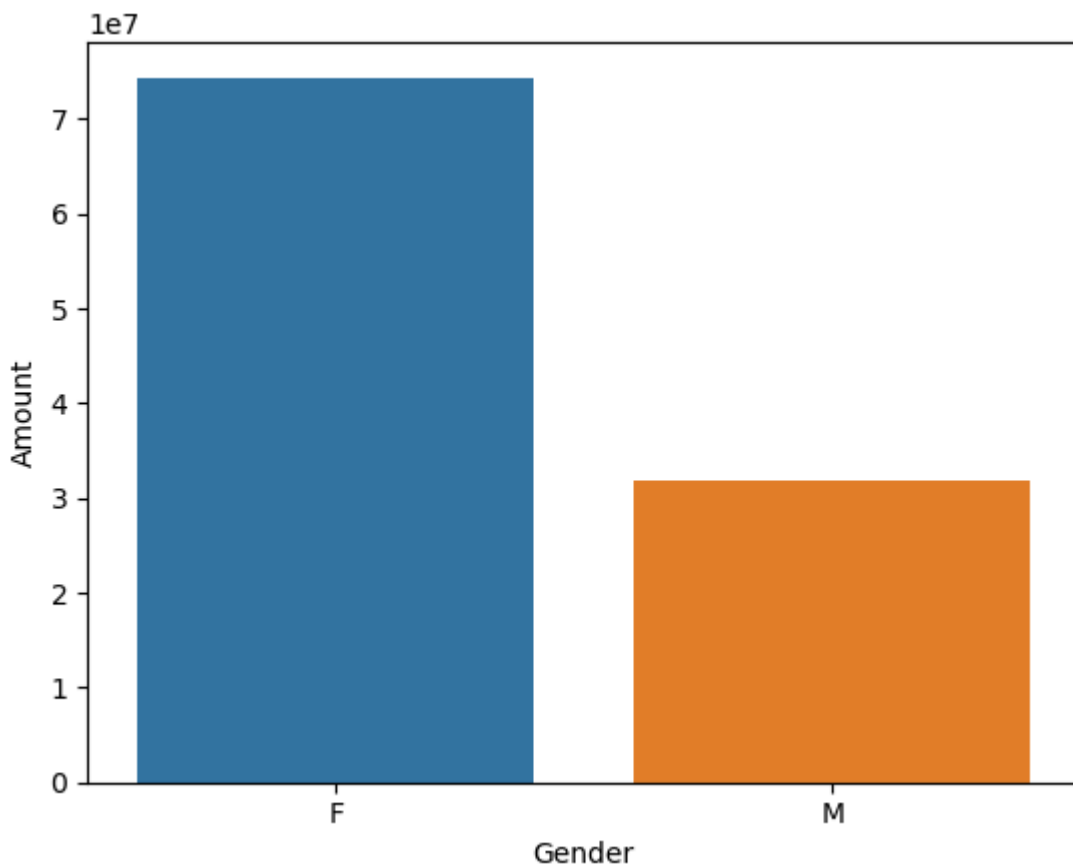


```
In [33]: # plotting a bar chart for gender vs total amount

sales_gen = df.groupby(['Gender'], as_index=False)['Amount'].sum().sort_values(by='Amount', a

sns.barplot(x = 'Gender',y= 'Amount' ,data = sales_gen)

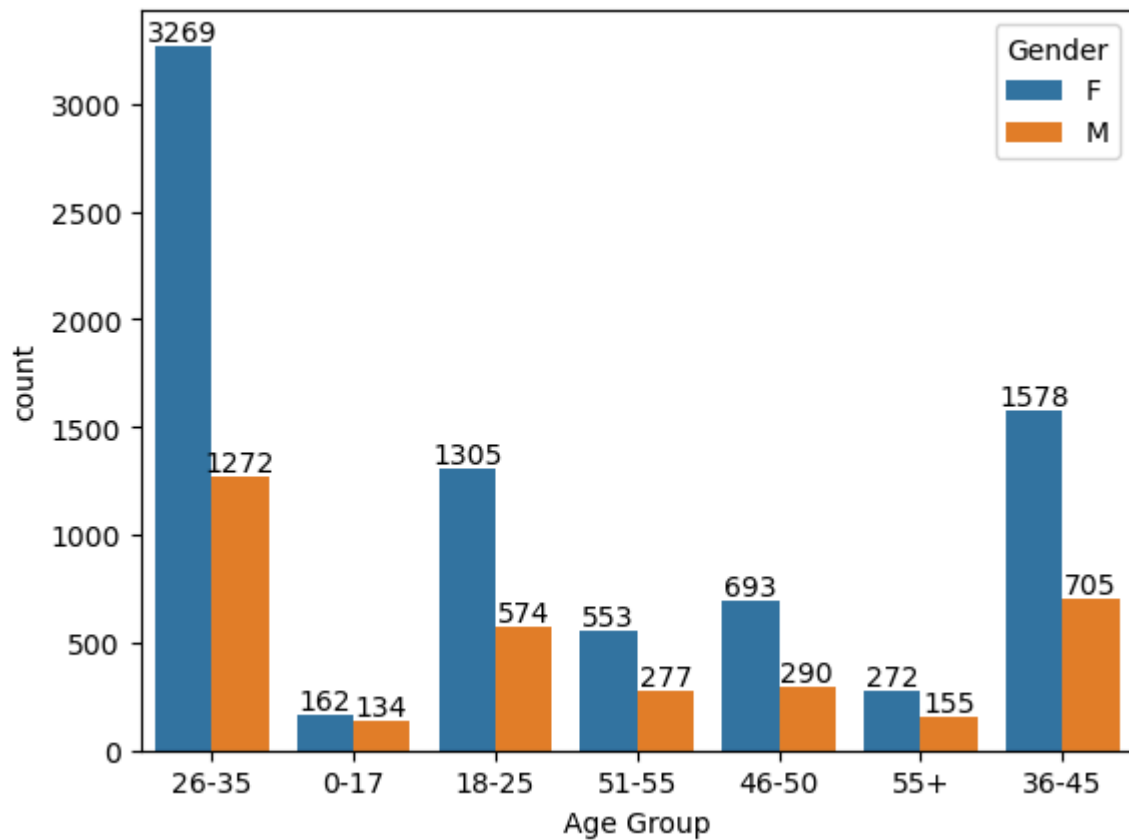
Out[33]: <AxesSubplot:xlabel='Gender', ylabel='Amount'>
```



From above graphs we can see that most of the buyers are females and even the purchasing power of females are greater than men

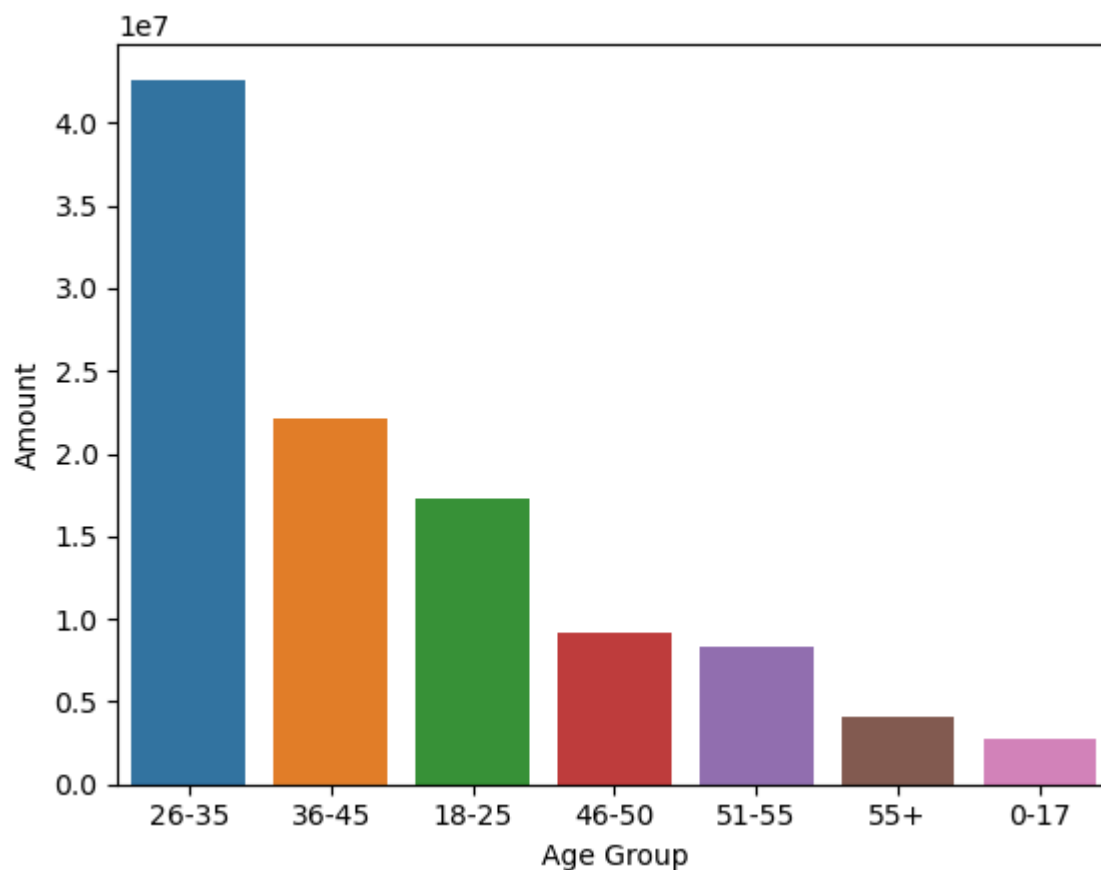
Age

```
In [34]: ax = sns.countplot(data = df, x = 'Age Group', hue = 'Gender')  
  
for bars in ax.containers:  
    ax.bar_label(bars)
```



```
In [35]: # Total Amount vs Age Group  
sales_age = df.groupby(['Age Group'], as_index=False)['Amount'].sum().sort_values(by='Amount')  
  
sns.barplot(x = 'Age Group', y = 'Amount', data = sales_age)
```

```
Out[35]: <AxesSubplot:xlabel='Age Group', ylabel='Amount'>
```



From above graphs we can see that most of the buyers are of age group between 26-35 yrs female

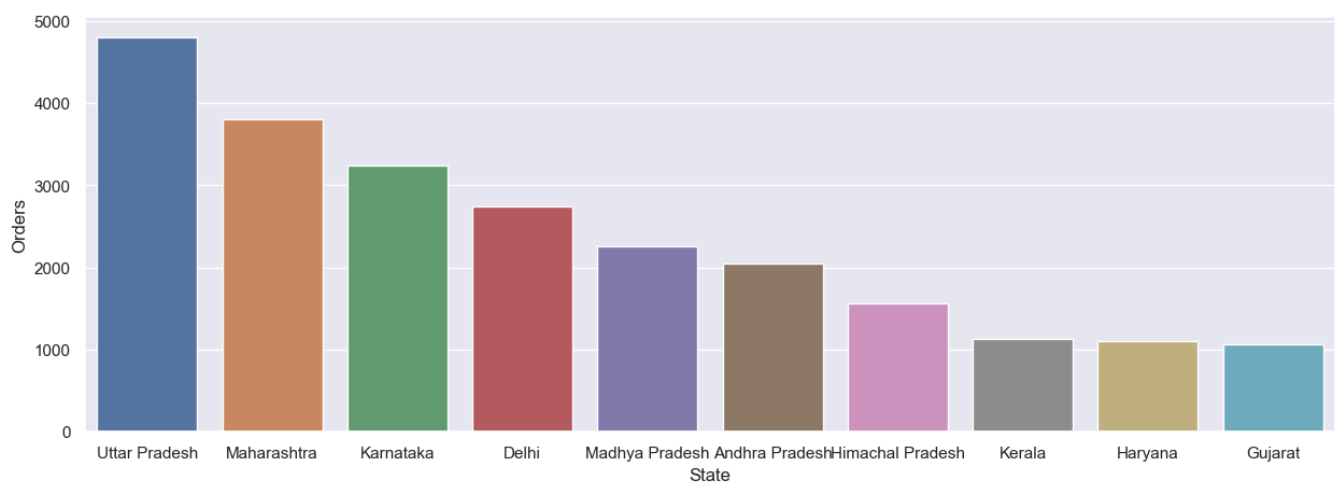
State

```
In [36]: # total number of orders from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Orders'].sum().sort_values(by='Orders',

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Orders')
```

Out[36]: <AxesSubplot:xlabel='State', ylabel='Orders'>

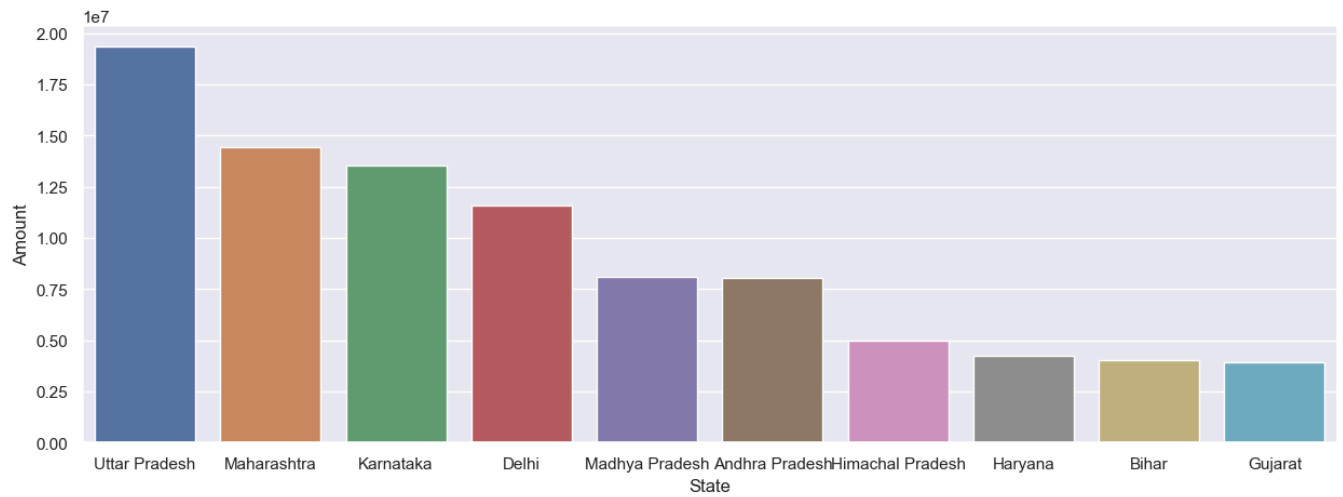


```
In [37]: # total amount/sales from top 10 states

sales_state = df.groupby(['State'], as_index=False)['Amount'].sum().sort_values(by='Amount',

sns.set(rc={'figure.figsize':(15,5)})
sns.barplot(data = sales_state, x = 'State',y= 'Amount')
```

Out[37]: <AxesSubplot:xlabel='State', ylabel='Amount'>

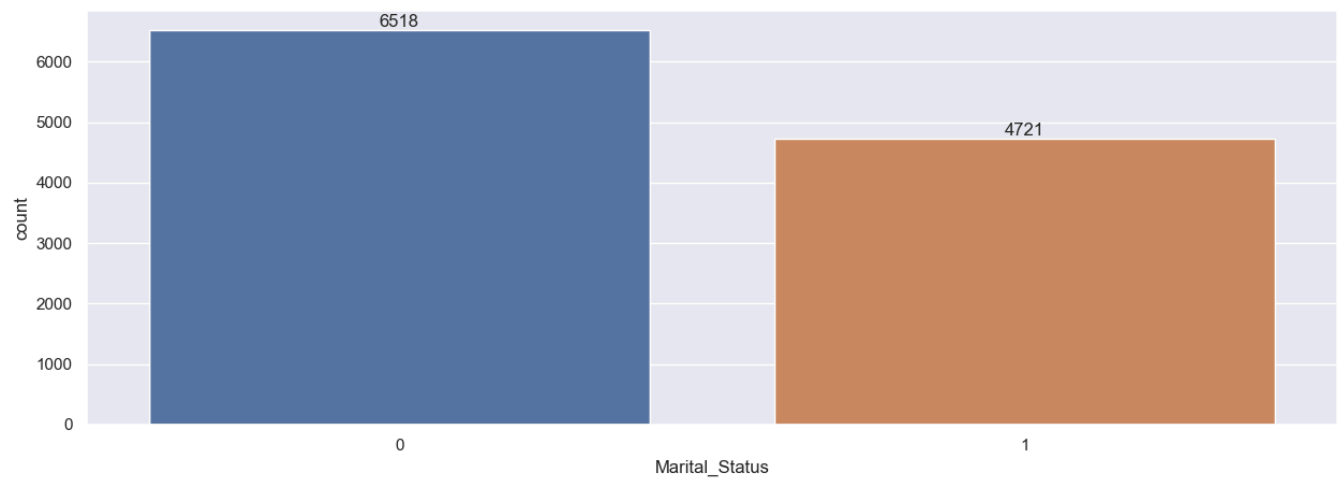


From above graphs we can see that most of the orders & total sales/amount are from Uttar Pradesh, Maharashtra and Karnataka respectively

Marital Status

```
In [38]: ax = sns.countplot(data = df, x = 'Marital_Status')

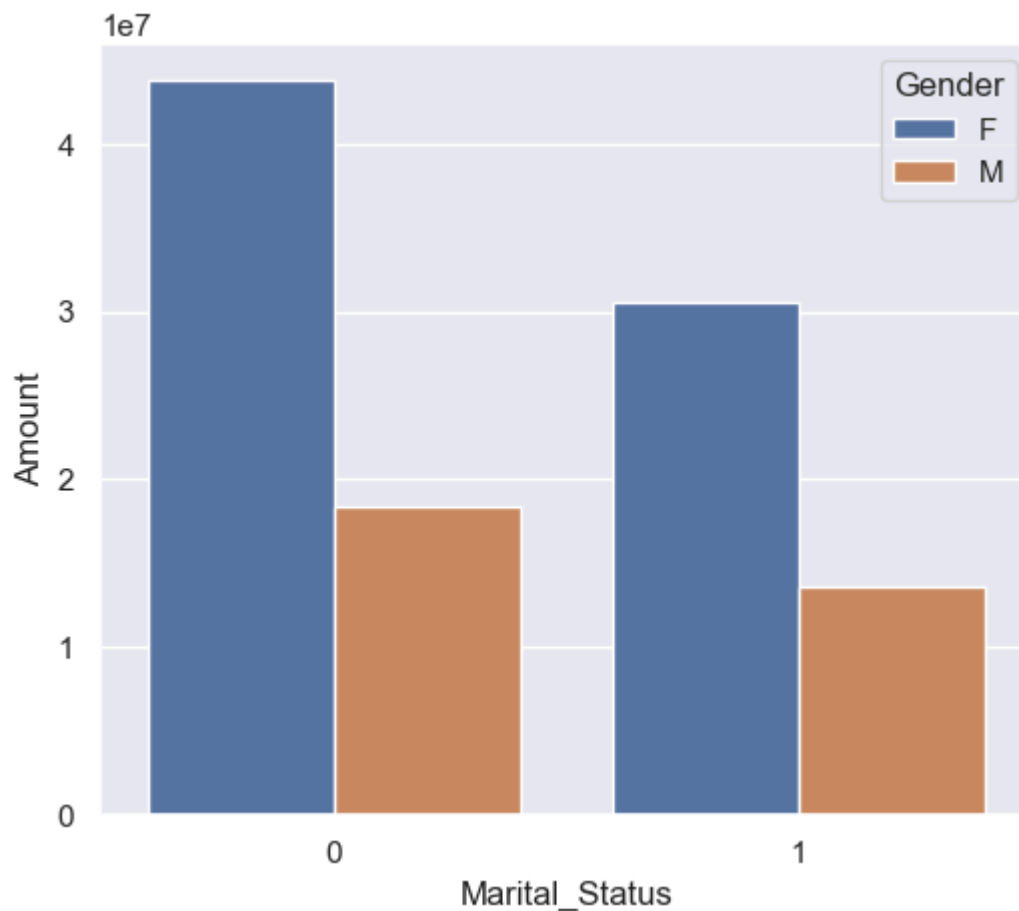
sns.set(rc={'figure.figsize':(7,5)})
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [39]: sales_state = df.groupby(['Marital_Status', 'Gender'], as_index=False)['Amount'].sum().sort_v

sns.set(rc={'figure.figsize':(6,5)})
sns.barplot(data = sales_state, x = 'Marital_Status', y= 'Amount', hue='Gender')
```

Out[39]: <AxesSubplot:xlabel='Marital_Status', ylabel='Amount'>

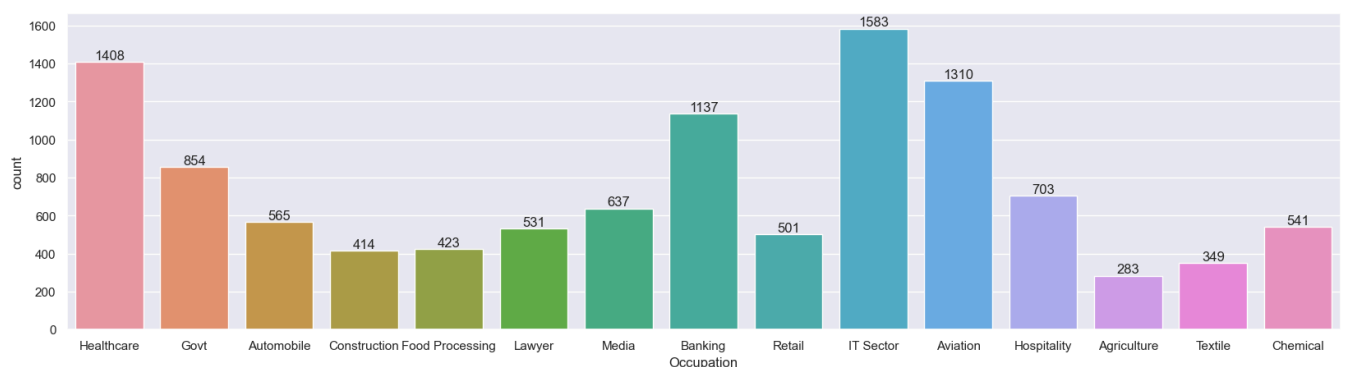


From above graphs we can see that most of the buyers are married (women) and they have high purchasing power

Occupation

```
In [40]: sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Occupation')

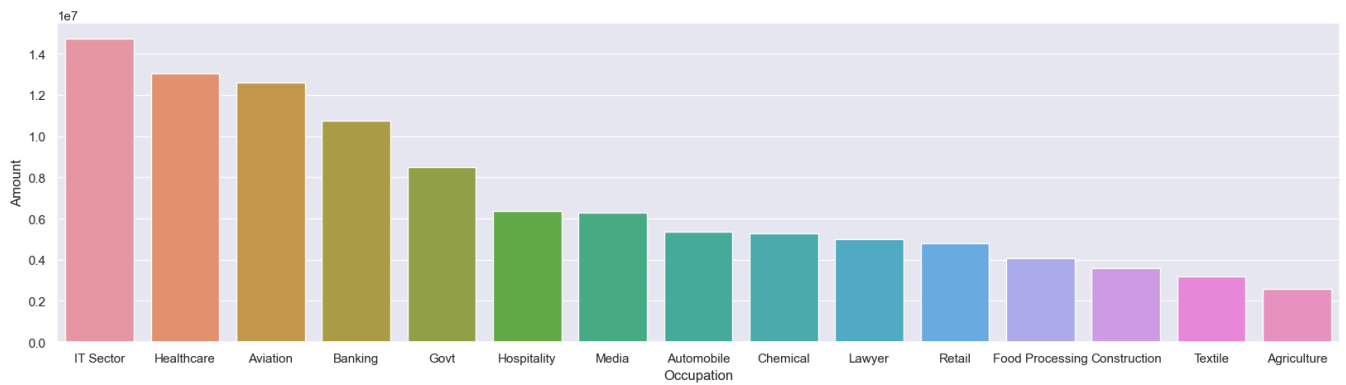
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [41]: sales_state = df.groupby(['Occupation'], as_index=False)['Amount'].sum().sort_values(by='Amount')

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Occupation', y = 'Amount')
```

```
Out[41]: <AxesSubplot:xlabel='Occupation', ylabel='Amount'>
```

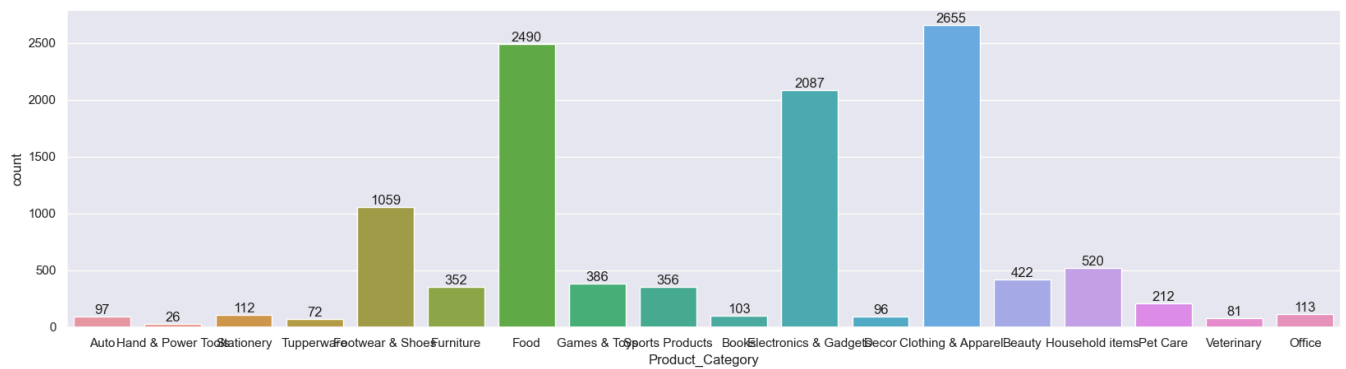


From above graphs we can see that most of the buyers are working in IT, Healthcare and Aviation sector

Product Category

```
In [42]: sns.set(rc={'figure.figsize':(20,5)})
ax = sns.countplot(data = df, x = 'Product_Category')

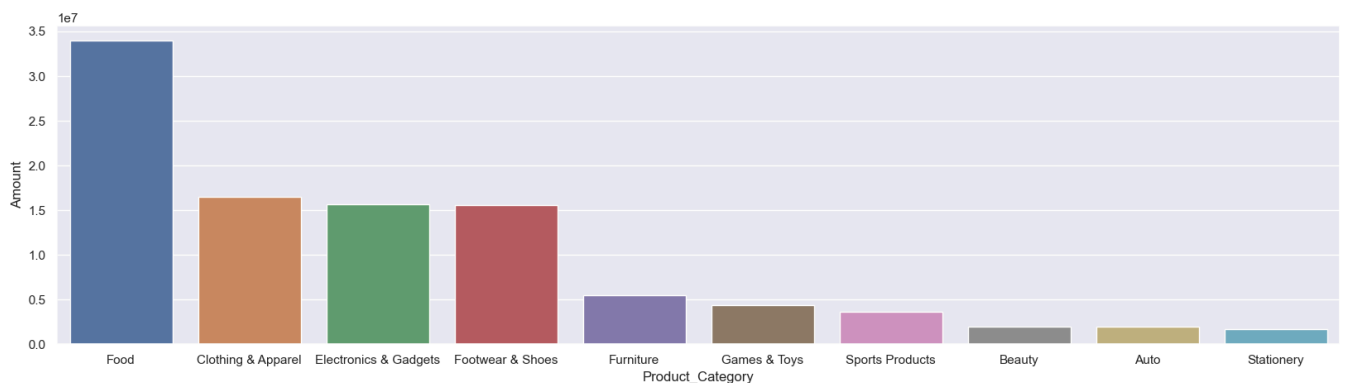
for bars in ax.containers:
    ax.bar_label(bars)
```



```
In [43]: sales_state = df.groupby(['Product_Category'], as_index=False)['Amount'].sum().sort_values(by=

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_Category',y= 'Amount')
```

```
Out[43]: <AxesSubplot:xlabel='Product_Category', ylabel='Amount'>
```

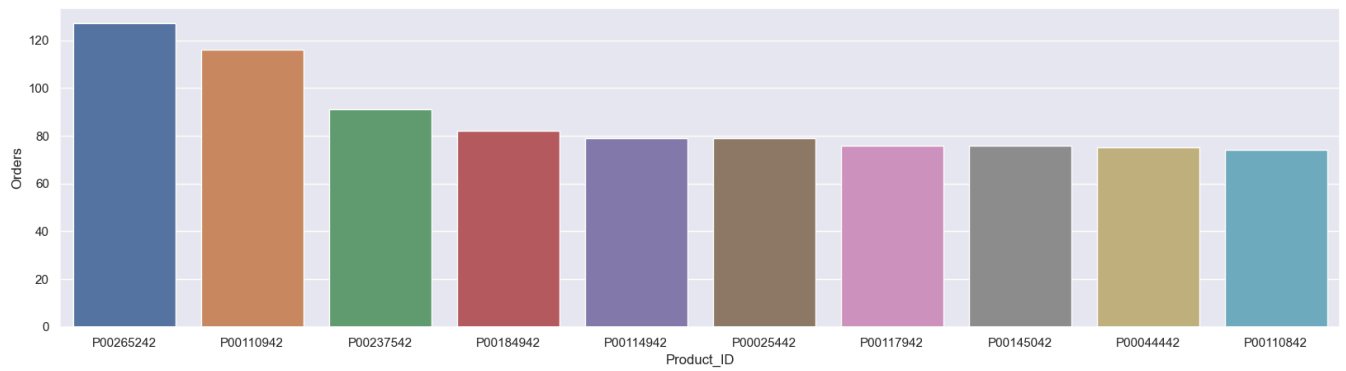


From above graphs we can see that most of the sold products are from Food, Clothing and Electronics category

```
In [44]: sales_state = df.groupby(['Product_ID'], as_index=False)['Orders'].sum().sort_values(by='Orders')

sns.set(rc={'figure.figsize':(20,5)})
sns.barplot(data = sales_state, x = 'Product_ID',y= 'Orders')
```

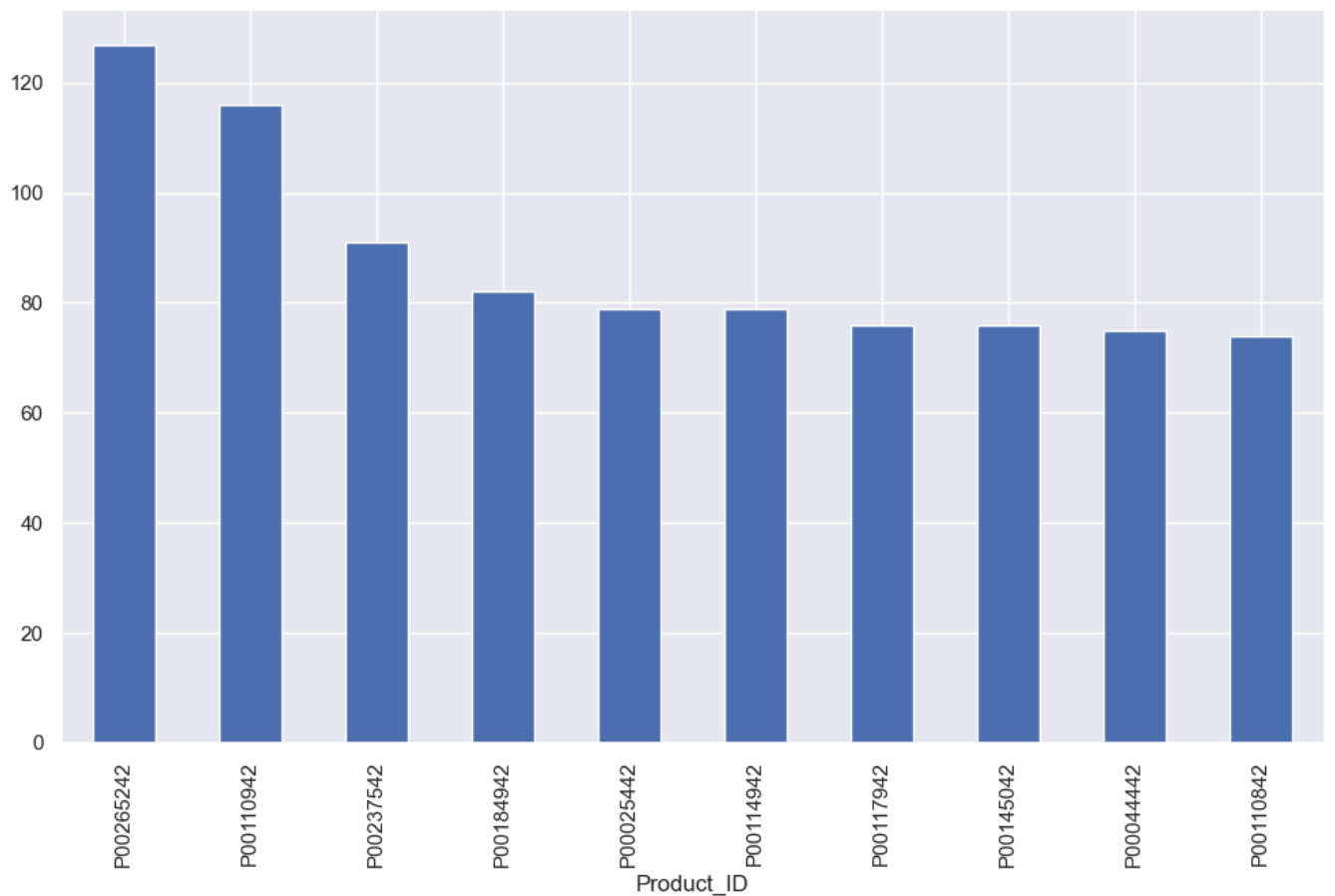
Out[44]: <AxesSubplot:xlabel='Product_ID', ylabel='Orders'>



In [45]: *# top 10 most sold products (same thing as above)*

```
fig1, ax1 = plt.subplots(figsize=(12,7))  
df.groupby('Product_ID')['Orders'].sum().nlargest(10).sort_values(ascending=False).plot(kind=
```

Out[45]: <AxesSubplot:xlabel='Product_ID'>



Conclusion:

Married women age group 26-35 yrs from UP, Maharastra and Karnataka working in IT, Healthcare and Aviation are more likely to buy products from Food, Clothing and Electronics category